Aptech Computer Education Examination

Testing Android Applications - Modular Exam Sample 2

Exam Code: MOD-TEST-AND-402 | Total Questions: 20 | Total Marks: 20.00 | Passing Marks: 8.00

Question 1: What is the default execution timeout for test runs on AWS Device Farm per device?

a) **60 minutes**
b) 40 minutes
c) 50 minutes
d) 90 minutes

Reference / Note: AWS Device Farm sets a default test execution timeout of 60 minutes per device run.

Question 2: While interacting with live devices during remote testing in AWS Device Farm, what does Device Farm do?

a) **Displays the device's real-time screen on the browser and accepts input gestures**
b) Limits human interaction to automated log downloads only
c) Cancels the active test run when mouse clicks are detected
d) Restarts the device into bootloader recovery mode

Reference / Note: AWS Device Farm remote access streams video of the physical device screen to a web browser window.

Question 3: Which of the following is a primary advantage of using Calabash for mobile test automation?

a) **Improves software quality and increases predictability through automated Cucumber acceptance tests**
b) Eliminates the requirement for writing application code
c) Bypasses all app permission checks automatically
d) Accelerates device battery charging during test runs

Reference / Note: Calabash enables writing and executing automated acceptance tests across iOS and Android using Cucumber specifications.

Question 4: In Android graphics and UI rendering performance profiling, what is the target frame rate threshold for smooth UI rendering?

a) **60 frames per second (16.6 ms/frame)**
b) 30 frames per second
c) 120 frames captured in profile buffer
d) 90 frames per second

Reference / Note: Android targets 60 fps (16 milliseconds per frame) for fluid rendering without jank or dropped frames.

Question 5: Which automated testing tool works on both real and virtual devices using stable and unique object recognition methods?

a) **Experitest (SeeTest) / Froglogic Squish**
b) Sublime Text
c) JProfiler
d) WireShark

Reference / Note: Experitest SeeTest and Froglogic Squish specialize in cross-platform GUI test automation using object properties.

Question 6: Which component in the Android platform facilitates the interaction of Java/Kotlin application code with native C/C++ libraries?

a) **JNI (Java Native Interface)**
b) ABI (Application Binary Interface)
c) Dalvik Bytecode
d) DEX compiler

Reference / Note: The Java Native Interface (JNI) defines a way for code written in Java to interact with native code written in C/C++.

Question 7: Which Android testing class is specifically designed for isolated testing of a single ContentProvider with a MockContentResolver?

a) **ProviderTestCase2**
b) ServiceTestCase
c) ApplicationTestCase
d) ActivityInstrumentationTestCase2

Reference / Note: Aptech Semester 3: Android Application Testing, Session 2, Page 40 & Session Check Question 5, Page 73: 'ProviderTestCase2 is a class used for isolated testing of a single ContentProvider.'

Question 8: Which JUnit assert method verifies that a specified boolean condition evaluates to true?

a) **assertTrue()**
b) assertCondition()
c) assertValid()
d) assertSame()

Reference / Note: JUnit's `assertTrue(boolean condition)` asserts that a condition is true; if not, it throws an AssertionError.

Question 9: Which Android testing class is specifically designed for isolated unit testing of a single ContentProvider with a mock environment?

a) **ProviderTestCase2**
b) MockContentResolver
c) IsolatedContext
d) ActivityInstrumentationTestCase2

Reference / Note: ProviderTestCase2<T> tests a single ContentProvider in isolation, using an IsolatedContext and MockContentResolver.

Question 10: Which InstrumentationTestCase helper methods are used to simulate user typing or interaction with a hardware keyboard or DPAD?

a) **sendKeys() and sendRepeatedKeys()**
b) typeText() and tap()
c) pressKey() and releaseKey()
d) dispatchInput() and flush()

Reference / Note: InstrumentationTestCase provides `sendKeys()` and `sendRepeatedKeys()` to send simulated key sequences to the focused view.

Question 11: Which assertion method verifies whether a given collection contains expected elements regardless of their sequence?

a) **assertContainsInAnyOrder()**
b) assertContainsInOrder()

c) assertSequenceEquals()

d) assertMatchesRegex()

Reference / Note: Matcher/assertion libraries like Hamcrest and MoreAsserts provide `assertContainsInAnyOrder()` to check items regardless of order.

Question 12: Which TouchUtils method is invoked in instrumentation tests to drag the screen upwards from the center by one quarter of the display height?

a) **dragQuarterScreenUp(InstrumentationTestCase test, Activity activity)**

b) dragQuarterScreenDown(InstrumentationTestCase test, Activity activity)

c) dragViewToBottom(InstrumentationTestCase test, Activity activity, View v)

d) scrollDownQuarterScreen(Activity activity)

Reference / Note: TouchUtils.dragQuarterScreenUp(test, activity) simulates a swipe from the center dragging up by 25% of the screen.

Question 13: A _____ is an Android project structure that contains test case source code and a test manifest targeting an application package.

a) **Test Project**

b) Library Project

c) Application Module

d) AIDL Bundle

Reference / Note: In classic Android tooling, a Test Project is created specifically to house test source code targeting the main application.

Question 14: Which XML tag in AndroidManifest.xml is used to register a broadcast receiver component?

a) **<receiver>**

b) <service>

c) <provider>

d) <activity>

Reference / Note: Broadcast receivers are declared in the manifest using the `<receiver android:name="...">` tag.

Question 15: Which method in a JUnit test fixture is executed before each test method to initialize test preconditions?

a) **setUp()**

b) tearDown()

c) testPreconditions()

d) runBare()

Reference / Note: In JUnit 3/4 test fixtures, setUp() is called before every test execution to prepare the testing environment.

Question 16: When creating an Android test project via command line tools, which argument specifies the directory location where the project is saved?

a) **--path <path-to-workspace>**

b) --target-id

c) --name <project-name>

d) --package <package-name>

Reference / Note: The `android create test-project` command uses the `--path` parameter to specify the workspace location.

Question 17: Which of the following is NOT an action available in Android Studio's Run/Debug Configuration dialog?

a) **Defecting a configuration set**
b) Creating a new run/debug configuration
c) Editing an existing configuration
d) Sorting and grouping configurations

Reference / Note: Run/Debug Configurations can be created, edited, copied, sorted, and deleted; 'defecting' is an invalid option.

Question 18: Which single archive file package contains compiled DEX bytecode, assets, resources, certificates, and manifest for deployment to Android devices?

a) **APK (Android Package)**
b) AAR (Android Archive)
c) JAR file
d) OBB (Opaque Binary Blob)

Reference / Note: An Android Package (.apk) file contains everything required to install and execute an app on Android.

Question 19: Which method in Android testing verifies that an Activity component enforces an explicit permission check at runtime?

a) **assertActivityRequiresPermission(String pkg, String activity, String permission)**
b) checkPermissionGranted(String permission)
c) verifySecuritySandbox(Context context)
d) assertPermissionAllowed(String permission)

Reference / Note: Android testing frameworks provide `assertActivityRequiresPermission` to ensure components enforce declared permissions.

Question 20: Which mock class provides an isolated mock execution context that isolates file operations and database queries from the real system during tests?

a) **IsolatedContext**
b) MockContext
c) RenamingDelegatingContext
d) ContextWrapper

Reference / Note: IsolatedContext delegates to a Context while shielding the real Android system by returning stub/mock system services.